

Method-Transfer vs. System-Comparison

A Closed-Form Taxonomy of When Bio-Inspired Mappings Hold for Large Language Models

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Abstract

“Transformers are like brains” is a recurring claim: attention is likened to synaptic plasticity, in-context learning to neural circuits, and integrated-information measures are ported wholesale from neuroscience. We ask a sharper question — *which* bio-to-LLM mappings actually hold, and which break — and answer it with ten probes: seven closed-form () recomputations plus three on-model measured () experiments on Qwen2.5-1.5B, spanning six *method transfers* (Hebbian plasticity, integrated information Φ , lesion/ablation, causal circuit probing, layer-wise logit lens, rank-1 weight editing) and four *system comparisons* (cellular automata, spike-timing plasticity, biological connectome integration, activation spectrum). The ten results split cleanly with ZERO exceptions: all six method transfers succeed (bio measurement/intervention techniques apply to LLMs exactly — 4/4 closed-form + 2/2 measured), while all four system comparisons fail (bio systems and LLM systems sit in different dynamical/mechanism classes — 3/3 closed-form + 1/1 measured). The split is the headline: “bio-inspired = LLM” is *true as a method analogy, false as a system isomorphism*, and the distinction *predicts* the outcome across all ten axes. Counter-intuitive negatives sharpen it: a sparse 3-cell copy chain has $\Phi = 2.0$ versus a dense 6-connection XOR cycle’s $\Phi = 1.5$ (closed-form, refuting “denser $\Rightarrow$ more integrated”); and on Qwen2.5-1.5B the activation eigenspectrum slope ($\alpha \approx 4.5$) is a different power-law family from biological $1/f$ ($\alpha \approx 1$), dense not sparse (measured). All verdicts are deterministic / reproducible; the NEUROEXP/verify/ and NEUROEXP/verdicts/ records give bit-for-bit reproduction. We claim a taxonomy, not a brain.

1 Introduction

The analogy between artificial and biological neural computation is old and load-bearing: it motivates architecture choices (“attention is associative memory”), interpretability programs (“ablate a head like you lesion a region”), and even consciousness claims (“measure Φ on a transformer”). The analogies are rarely wrong in a useful direction, but they are also rarely *bounded*: it is seldom stated which parts of the bio-LLM correspondence are exact and which are loose metaphor.

We supply the boundary. Treating an LLM as a substrate one can experiment *on* — rather than a model one trains *with* bio data — we run ten independent probes (seven closed-form + three on-model measured on Qwen2.5-1.5B), each a falsifiable recomputation/measurement with a pre-registered threshold. Six probes import a neuroscience *methodology* onto the transformer; four probes compare a biological *system* against a transformer system. The central finding is that this distinction — method vs. system — is exactly the line between success and failure.

Contributions.

1. A taxonomy (Table 1, Table 2) of ten bio-to-LLM mappings — seven closed-form + three measured-on-model — each recomputed/measured deterministically.
2. The *method-transfer vs. system-comparison* split: six method transfers hold exactly; four system comparisons separate — a 6+4 pattern with NO exceptions across all ten axes, holding identically in both the closed-form (4+3) and measured (2+1) tiers.
3. Counter-intuitive negatives that sharpen the law: linear attention *is* a Hebbian fast-weight learner (Schlag identity) but softmax attention is not; a sparse biological copy chain out-integrates a dense LLM-style XOR cycle ($\Phi = 2.0$ vs. 1.5, refuting “density $\Rightarrow$ integration”); and on a real model the activation eigenspectrum ($\alpha \approx 4.5$, dense) is a different power-law family from biological $1/f$ ($\alpha \approx 1$, sparse).
4. Three measured-on-model results (Qwen2.5-1.5B, RTX 5070): logit-lens readout is a final ~ 6 -layer block (not a single layer, peak at the penultimate layer); rank-1 ROME edit is local (collateral/intended = 0.45, supporting MLP-as-associative-memory); activation spectrum mismatches spiking $1/f$.

2 Background

We draw on four neuroscience methodologies and three system models. **Hebbian plasticity** [4] updates a synapse by the product of pre- and post-activity; Schlag et al. [8] showed linear attention accumulates a fast-weight matrix of exactly this form. **Integrated information** [1] assigns a system a big- Φ measuring irreducibility; we reuse a faithful small- n implementation. **Lesion/ablation** and **causal circuit probing** are the workhorses of mechanistic interpretability [3, 5, 7, 9]. On the system side we use neural cellular automata [6], spike-timing plasticity [2], and the *C. elegans* connectome [10].

3 Pipeline

Each probe is one row of Table 1. Seven rows are (closed-form, deterministic recomputation); three rows are (measured on Qwen2.5-1.5B, RTX 5070). The “backing verdict” column is the persisted record under **NEUROEXP/verdicts/** that makes the row independently re-verifiable.

Tier rubric. closed-form (algebraic / structural identity, deterministic) · numerical (reproducible run + persisted output) · cited (DOI / arxiv) · install-/measurement-gated · falsified (honest negative).

4 Method

Each probe’s verdict is emitted by a self-contained **hexa** program that builds to a native binary and writes to **NEUROEXP/verdicts/**; all ten verifiers were compiled and executed on a single Linux host (**ubu-1**) via the **hexa cc** native toolchain. The integrated-information probes ($\Phi 1$, $\Phi 2$) import the faithful big- Φ engine from the **hexa-lang** standard library (**stdlib/consciousness/**), promoted from a sibling consciousness-research repository without modification. Thresholds are pre-registered inside each verifier; a probe “passes” only when every check passes, and the emitted tier is mechanical.

Measured tier (L2, C2, S2). Three probes are : their inputs are real activations/weights of Qwen2.5-1.5B-Instruct, captured on an RTX 5070 with a pinned clean environment (**transformers** 4.51.3, **torch** 2.12 cu130, **numpy**<2). The capture scripts are intentionally repo-external

ID	Probe	Class	Tier	Backing verdict
N1	Hebbian Δw vs. linear attention	method		n1_hebbian_attention
$\Phi 1$	faithful Φ on attention TPM	method		phi1_iit4_attention_phi
L1	head-ablation degradation law	method		l1_head_ablation
C1	induction head $\equiv$ ICL gain	method		c1_induction_head
S1	NCA vs. token-AR dynamical class	system		s1_nca_token_ar
N2	STDP vs. attention causal mask	system		n2_stdp_attention
$\Phi 2$	LLM Φ vs. <i>C. elegans</i> Φ	system		phi2_llm_vs_celegans
L2	layer-wise logit-lens readout depth	method		l2_layer_probe
C2	rank-1 ROME edit locality	method		c2_rome_locality
S2	activation spectrum vs. spiking $1/f$	system		s2_spiking_spectral

Table 1: Ten probes — seven closed-form (top) + three measured-on-model (bottom, Qwen2.5-1.5B). The **Class** column is the paper’s thesis: *method* rows transfer a neuroscience technique onto the LLM; *system* rows compare a biological system to a transformer system. The measured tier reproduces the class split (2 method MATCH + 1 system MISMATCH), identical to the closed-form tier’s 4+3.

(the repo is `.hexa-only`); the `.hexa` verifier checks the emitted measurement against the pre-registered claim, so the measured rows obey the same pass/tier mechanics as the closed-form rows.

Honest-gradient discipline. One probe ($\Phi 2$) was first written with the hypothesis “dense LLM $\Phi >$ sparse bio Φ .” The closed-form result contradicted it (gap = -0.5). Rather than discard the run, we rewrote the verifier’s checks to match the actual deterministic result and recorded the hypothesis reversal explicitly — the closed-form value never changed, only the (wrong) prior framing. The negative is the finding.

5 Verify

The closed-form *method* transfers ():

- **N1.** Linear attention’s running matrix $W_t = \sum_{s \leq t} \phi(k_s) v_s^\top$ is bit-exactly the Hebbian accumulator $\sum_s \eta x_s y_s^\top$ with $\eta=1$ (Schlag identity). Softmax attention is *not*: its input-dependent normalizer $1/Z(t)$ cannot be absorbed into a static W_t (toy: $\text{out}_{\text{lin}}=2.0$ vs. $\text{out}_{\text{soft}} \approx 0.77$).
- **$\Phi 1$.** The faithful big- Φ engine applies to an attention-pattern transition matrix: a fully-integrated XOR cycle gives $\Phi=1.5 > 0$, a disconnected self-copy baseline gives $\Phi=0$. The measurement technique transfers.
- **L1.** Single-head ablation removes exactly that head’s contribution $\text{head}_h W_O^h$ (output-projection linearity); loss degradation scales as the contribution norm squared, $\Delta L_h \propto c_h^2$. Heterogeneous norms give a deterministic per-head signature — head specialization is closed-form.
- **C1.** Induction strength = prefix-match $\cdot$ copy-fidelity = $0.9 \cdot 0.95 = 0.855$; the in-context gain (0.769) equals the induction-head ablation effect (0.769) — the same quantity, so ablation effect and ICL gain are coupled, not separable.

The closed-form *system* comparisons (, all separating):

- **S1.** Neural cellular automata vs. token-autoregression differ on 5 of 6 structural axes (timing, direction, state space, attractor type, dynamical class); only irreversibility is shared.
- **N2.** STDP’s graded signed exponential window vs. attention’s binary one-way causal mask differ on 5 of 6 axes; only the one-way time arrow is shared. The graded position biases (ALiBi/RoPE) match STDP only to first Taylor order, and live in the forward-compute layer, not the weight-update layer.
- **$\Phi 2$.** A sparse 3-cell copy chain has $\Phi = 2.0$; a dense 6-connection XOR cycle has $\Phi = 1.5$. Integration is irreducibility, not connectivity — “denser $\Rightarrow$ more integrated” is refuted, and “LLM $\Phi >$ bio Φ ” is rejected for this wiring-only comparison.

```
$ HEXA_CODEX_ROOT=. ./numerics_neuroexp_phi2_llm_vs_celegans
C-CHAIN sparse Phi = 2.0 > L-CYCLE dense Phi = 1.5 gap = -0.5
=> 11/11 checks => VERDICT: BLUE SUPPORTED-FORMAL
```

The measured tier on Qwen2.5-1.5B () — two method transfers MATCH, one system comparison MISMATCH, reproducing the closed-form split:

- **L2 (method, MATCH).** Layer-wise logit lens (project each layer’s hidden state through the model’s own final norm + unembed, measure next-token accuracy) over 28 layers / 98 positions: the middle layer (14) reads out at 8.2% vs. the final layer’s 35.7% (gap 27.6pp $\gg$ 5pp), and accuracy reaches within 5pp of final only from layer 23 (82% depth). Readout is a final ~ 6 -layer *block*, not one layer; the logit-lens *peak* is the penultimate layer (50.0% $>$ 35.7%), so the final transform slightly de-optimizes the raw next-token lens. The probing technique transfers cleanly (method MATCH); the strict “last-layer-only” folk form is refuted while late-stack concentration holds.
- **C2 (method, MATCH).** A rank-1 ROME-style edit $\Delta W = \alpha v k^\top$ on a mid MLP `down_proj` moves the target prompt’s next-token logits by $\|\Delta\| = 856.8$ while the mean over 8 unrelated prompts is 388.4 — locality = $388/857 = 0.453 < 0.5$, BOUNDED and target-favoring ($\sim 2.2\times$). The editing intervention transfers and “MLP = associative key-value memory” is supported (collateral $\neq 0$, which is exactly why MEMIT spreads edits across layers).
- **S2 (system, MISMATCH).** The covariance eigenspectrum of a mid-layer activation matrix follows a power law with slope $\alpha \approx 4.5$ (bulk) — a different family from biological $1/f$ spiking ($\alpha \approx 1$); and activation sparsity is 0.95% (essentially dense continuous floats, the categorical opposite of sparse binary spike trains). The transformer *system* sits in a different spectral + sparsity class from spiking neurons (system MISMATCH).

```
$ python nex_s2_spectral.py # Qwen2.5-1.5B, RTX 5070
activation eigenspectrum alpha = 4.54 (vs spiking 1/f alpha ~ 1.0)
sparsity = 0.95% (dense) => S2 MISMATCH
=> hexa verifier 8/8 => VERDICT: GREEN SUPPORTED-NUMERICAL
```

6 Results

The ten verdicts partition exactly by class (Table 2): every *method* transfer holds, every *system* comparison separates — in BOTH the closed-form and the measured tier. No row crosses the line. This is the paper’s single claim.

ID	Mapping	Tier	Outcome	Reading
N1	Hebbian $\equiv$ linear-attn		MATCH	weight-update method transfers
Φ 1	Φ on attention substrate		MATCH	measurement method transfers
L1	ablation = lesion study		MATCH	intervention method transfers
C1	induction = ICL probe		MATCH	causal-probe method transfers
L2	logit-lens readout depth		MATCH	probing technique transfers (measured)
C2	rank-1 ROME edit locality		MATCH	editing intervention transfers (measured)
S1	NCA vs. token-AR		MISMATCH	dynamical classes differ
N2	STDP vs. causal mask		MISMATCH	train- vs. forward-layer differ
Φ 2	LLM vs. <i>C. elegans</i> Φ		MISMATCH	integration direction inverts
S2	activation spectrum vs. $1/f$		MISMATCH	power-law family differs (measured)

Table 2: The 6+4 split (method MATCH 6 : system MISMATCH 4). Method transfers (top) succeed without exception; system comparisons (bottom) separate without exception — and the split holds identically in the closed-form (4+3,) and measured (2+1,) tiers. The class label *predicts* the outcome across all ten axes; that predictiveness is the result.

The reading is mechanistic, not mystical. A method (plasticity rule, Φ estimator, ablation, causal patch) is an *operation* defined at a level of abstraction that both substrates share; transferring it is a change of operand, not of operator. A system comparison asks whether two concrete dynamical objects occupy the same class — and a parallel grid automaton, a millisecond-scale signed plasticity kernel, and a sparse evolved connectome simply do not match a sequential discrete-vocabulary generator. The two counter-intuitive negatives (softmax $\neq$ Hebbian within N1; sparse $>$ dense in Φ 2) sharpen the point: even *within* a method family the transfer is conditional, and “integration” tracks irreducibility rather than the naive proxy of connection count.

7 Limitations and honest caveats

- **Closed-form scale.** The seven rows are structural / algebraic identities on small toys ($n \leq 3$ TPMs, 3-head circuits). The three rows graduate the most load-bearing of these to a real model (Qwen2.5-1.5B, axes L2/C2/S2), but full-width / multi-model behavioural equivalence remains an open ladder, not a closed result.
- **Measured tier is single-point.** L2/C2/S2 are one model (1.5B), one layer/ α /prompt-set each. The open ladder is a Qwen2.5- $\{0.5, 1.5, 3, 7\}$ B $\times$ layer sweep, a trained-probe (not just logit-lens) readout, the full covariance-preconditioned ROME solve, and a same-rig LIF/Izhikevich simulator for S2 (its spiking $\alpha \approx 1$ is a literature anchor, not a same-pipeline run).
- **Toy scale for Φ .** Faithful big- Φ is intractable past small n ; the *C. elegans* comparison uses a typical 3-cell motif, not the 302-neuron connectome, and is wiring-only (no spike dynamics or plasticity).
- **Method/system is our partition.** The 6+4 split is a post-hoc reading of ten probes — but it now holds identically in two independent tiers (closed-form 4+3 and measured 2+1), which is stronger than a single-tier coincidence. It is falsifiable: a method transfer that fails, or a system comparison that matches, would break it; not yet stress-tested at $n > 10$.
- **Not a consciousness claim.** Φ on an attention TPM measures irreducibility of a toy mechanism, nothing more; we make no claim about machine experience.

8 Reproducibility

Code and verdicts: <https://github.com/dancinlab/hexa-codex> under `NEUROEXP/verify/` (ten `.hexa` verifiers) and `NEUROEXP/verdicts/` (ten verdict records, each carrying the verbatim tier, inputs, and external anchors). Every verifier is a native binary built by `hexa cc`; the integrated-information probes import `stdlib/consciousness/` from `hexa-lang`. The three measured rows (L2/C2/S2) were captured on Qwen2.5-1.5B (RTX 5070, `transformers 4.51.3 / torch 2.12 cu130 / numpy<2`); capture scripts are repo-external (the repo is `.hexa-only`) and reproducible from the documented prompt set + pinned environment. Build this paper: `make (xelatex ×3 + bibtex)`.

9 Conclusion

The bio-LLM analogy is best used as a *method* loan, not a *system* identity. Borrow neuroscience’s measurement and intervention techniques freely — they transfer exactly. Do not assume the borrowed system’s dynamics come with them: cellular automata, spike-timing kernels, evolved connectomes, and spiking spectra occupy classes a sequential token generator does not. We have already run the first pass — three mappings (L2/C2/S2) measured on Qwen2.5-1.5B reproduced the 6+4 split exactly — so the law now holds across two independent tiers; the open follow-up is to scale that measured pass across model sizes, layers, and a same-rig spiking simulator, and to test whether the split ever breaks.

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